Separating reconstruction, generation and antibacterial activity in antimicrobial peptide design

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Abstract

Antimicrobial peptide design combines sequence reconstruction, de novo generation and experimental activity assessment, but each endpoint answers a different evaluation question. We present an endpoint-specific evaluation protocol that integrates a controlled reconstruction-objective comparison with sequence-distribution analysis and a separate antibacterial assay panel. Using 279 training and 41 development sequences, we compare categorical (CE21) and squared-error (MSE21) objectives under matched architecture, initialization, batches and decoding. CE21 increases development residue micro accuracy from 12.55% to 70.90% and improves six distributional distances across both terminal conditions and both reference sets. A terminal-conditioned LSTM has the lowest values on all 24 specified distance comparisons, demonstrating the additional information obtained from direct generation evaluation. In a separate panel of 27 experimentally tested peptides, six inhibit *Staphylococcus aureus* ATCC 29213 at MICs $\leq$31.25 µg/mL, and seven have MICs $<$200 µg/mL against at least one of four reference strains. These activity counts are consistent across the original MIC summary and a replicate-based sensitivity analysis. The protocol connects each conclusion to its measured endpoint and provides a practical basis for interpreting reconstruction gains, sampled-sequence properties and antibacterial activity in peptide-design studies.

Keywords antimicrobial peptides, generative models, evaluation, variational autoencoder, diffusion models, antibacterial activity

Author description

The authors study machine learning for antimicrobial peptide discovery across computer science, industrial artificial intelligence and biotechnology, emphasizing reproducible evaluation of computational design and laboratory measurements.

Introduction

Antimicrobial peptides provide diverse sequence templates for antibacterial discovery, with activity shaped by sequence, structure and experimental context (1; 2). Recurrent models (3), variational autoencoders (4; 5; 6), conditional adversarial models (7; 8) and diffusion approaches (9; 10; 11) offer different ways to explore this sequence space. Integrated frameworks also combine generation, identification, activity prediction and optimization (12).

Computational design becomes biologically informative when generated or screened sequences are connected to explicit functional measurements. Studies combining generative models with molecular simulations (13; 14), large-scale sequence screening (15) and staged selection followed by assays, as in AMPGen (9), illustrate several routes to experimental candidates. These workflows contain distinct generation, selection and validation steps, a distinction also emphasized in recent synthesis of the field (16).

The evaluation question is whether an improvement at one step supports a conclusion about another. Reconstruction measures recovery of a known sequence from its encoded representation; de novo generation uses inputs available without that sequence. Generation benchmarks therefore use complementary statistics rather than a single score, as established for molecular design (17; 18). For biological sequences, residue composition, local sequence composition and length offer interpretable but incomplete views of sampled outputs (19; 20). MIC measurements answer a further question: whether a particular peptide inhibits a specified organism at a measured concentration. None of these observations alone determines the others.

AMP prediction studies show why the comparison design matters. Independent assessments have examined predictor performance (21), the benefit of deep versus shallow models (22), and changes in apparent performance caused by negative-data selection (23). Homology-aware partitioning and leakage control address a related dependence on dataset construction (24; 25). Biological machine-learning guidelines accordingly connect validation claims to the data partition, comparator and measured endpoint (26; 27; 28).

For peptide generators, recent comparative work examines model families and sampling choices (20), while standardized MIC and haemolytic-activity regression benchmarks address quantitative functional prediction (29). Species-aware models further emphasize the organism-specific nature of an activity endpoint (30). These contributions motivate a controlled assessment of whether improved residue recovery is accompanied by changes in sampled-sequence statistics, evaluated against an executable autoregressive reference.

We address this question by comparing categorical and squared-error objectives under matched VAE architecture, initialization, training schedule and decoding, then evaluating their sampled sequences alongside a terminal-conditioned LSTM. Latent perturbation analyses examine whether recoverable encoded information is sufficient for the observed sampling performance. A separate 27-peptide assay panel characterizes strain-specific antibacterial activity. The resulting protocol links each comparison to its inputs and endpoint: objective changes to reconstruction, sampled sequences to six explicit distances, and experimentally tested peptides to MIC measurements.

Materials and methods

Peptide dataset and terminal conditions

We constructed the computational cohort from 512 DBAASP detail cards (31). Eligibility required an unchanged canonical monomer sequence of 1–48 residues, an unmodified N terminus, a free or amidated C terminus, compatible molecular graphs and bacterial MIC metadata. We grouped sequences using global Needleman–Wunsch alignment (32) and an identity threshold of at least 70%, counting all alignment columns including gaps, and formed transitive connected components. Previously exposed components were assigned to training or excluded under the fixed exposure rule; remaining components received a seeded hash assignment with nominal 80/10/10 proportions. Descriptor scaling used training data only.

The comparison used 279 training sequences and 41 development sequences, including 701 non-padding development residues. The free and amidated groups contained 126 and 153 training sequences and 18 and 23 development sequences, respectively. Terminal state was the only externally supplied generation condition: generators received neither the true sequence, residue-aligned descriptors nor true length. During reconstruction, the encoder received sequence-derived descriptors as well as the sequence. Free and amidated groups contained different peptides and were analyzed separately.

VAE–diffusion models and reconstruction objectives

The categorical (CE21) and squared-error (MSE21) models used the variational autoencoder framework (33) and shared architecture, initialization, batches, random-number schedule, 48-position output, 21-category head and decoding rule. The categories were the 20 canonical amino acids and padding. Each VAE had 1,681,549 parameters; the associated diffusion model had 18,612,994 parameters. A third pipeline used scalar residue-number regression with a different output head and normalization and served as an additional pipeline comparator. Network layers and source configurations are specified in Supplementary Methods.

Let s_{ijc} denote the raw score for category c at position j of sample i, y_{ij} the target category including padding, and B the batch size. The two reconstruction objectives were

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{48B}\sum_{i}\sum_{j}\log \operatorname{softmax}(s_{ij})_{y_{ij}}$$

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{96B}\sum_{i}\sum_{j}\left\| s_{ij} - e_{y_{ij}} \right\|^2,$$

where e_y is the one-hot target. MSE was applied directly to raw scores with a factor of one half. Both objectives included padding positions and used the same optimizer, learning rate and update count. Following the denoising-diffusion framework (34), the diffusion model was trained on posterior parameters; generation sampled a latent representation and decoded it into a sequence. The KL coefficient was held fixed across the objective comparison.

Autoregressive comparator and training protocol

We trained a terminal-conditioned LSTM (35) with two recurrent layers of hidden width 64, token embeddings of width 32, terminal embeddings of width 8 and 62,501 parameters. Teacher-forced training and generation used an explicit end-of-sequence token and a 48-residue cap. The comparator used the same peptide data and evaluation metrics as the VAE pipelines; it was a local recurrent model rather than a reproduction of a published implementation (3).

Each VAE received 1,000 updates followed by 1,000 diffusion updates; the LSTM received 1,000 updates. We used one training seed per model and the final checkpoint, without checkpoint selection or hyperparameter search within this comparison. VAE and LSTM seeds were 222, the diffusion seed was 322, and generation seeds were 422 and 423 for free and amidated conditions. VAE and diffusion batch sizes were 32 and 8. Training and generation ceilings were 2,940 s and 1,800 s on one CPU thread; architectures therefore shared ceilings but differed in parameter count and realized computation. Each method generated 16 sequences per terminal condition, yielding 128 outputs across four methods. The development partition had been used during model development, so all model comparisons are descriptive. Full optimization settings and execution details appear in Supplementary Methods.

Reconstruction and generation evaluation

We evaluated reconstruction from the posterior mean. Residue micro accuracy used all 701 development residues, with missing predicted positions counted as incorrect; macro accuracy averaged recall over target-supported residue classes. Whole-sequence exact matches and length mean absolute error used 41 development sequences as the denominator.

We assessed generated sequences for canonical residues, length, uniqueness and exact training copies. Distributional comparisons retained every canonical output of length 1–48; eligibility at length 5–48 was recorded separately. We calculated six distances for each method and terminal condition against both training and development references: residue and dipeptide Jensen–Shannon divergence (36) in bits, the corresponding L_1 distances, length Kolmogorov–Smirnov distance and biased linear-kernel MMD^2 (37). The MMD feature vector contained length/48, 20 residue fractions and 400 dipeptide fractions, with each block weighted by $1/\sqrt{3}$. One-residue sequences had a zero dipeptide block. JS, L_1 and entropy used pooled counts; MMD averaged per-sequence features. Lower distances indicate

Table 1 Development reconstruction from the posterior mean. Counts and errors use 701 target residues or 41 sequences. The autoregressive model has no corresponding reconstruction task.

Model	Correct	Micro (%)	Macro (%)	Length MAE	Exact
CE21	497/701	70.90	42.76	0.000	3/41
MSE21	88/701	12.55	3.57	5.561	0/41
Scalar	111/701	15.83	8.05	1.341	0/41

closer agreement under the specified statistic. The linear-kernel MMD measures differences in these feature means; it does not identify the complete sequence distribution or constitute a hypothesis test. Generated sequences were not treated as independent training replicates.

Seven exploratory analyses examined latent-source replacement, coordinate permutation, a template-conditioned prior, KL warmup, latent-conditioned autoregression, direct-latent diffusion and conditioning-context variants. Each analysis used its specified reconstruction, information or generation criterion (Supplementary Methods; Table S4). Separate activity-regression analyses evaluated mean, ridge and convolutional predictors with source-group validation (Table S3). The convolutional predictor was an architecture adaptation from the cached QMAP implementation (29), associated with the regression approach of Witten and Witten (38); its data came from the separate DBAASP cache described in Supplementary Methods.

Antibacterial activity assays

We tested 27 candidate peptides against *Escherichia coli* ATCC 25922, *Pseudomonas aeruginosa* ATCC 27853, *Staphylococcus aureus* ATCC 29213 and *Enterococcus faecalis* ATCC 29212. This experimental panel was analyzed separately from the reconstruction-objective comparison. The assays used ten two-fold concentrations from 1000 to 1.953125 µg/mL, with three replicate wells per peptide, strain and concentration. Ampicillin, polymyxin B and daptomycin served as reference antibiotics. Sample sequences, purity measurements and the complete OD_{600} series are provided in Table S5. Concentration units, strain identity and replicate interpretation follow the endpoint-specific reporting principles of antimicrobial susceptibility testing (39; 40; 41).

We retained the original MIC summary and performed a replicate-based sensitivity analysis. In this analysis, a replicate endpoint was the lowest tested concentration with $OD_{600} < 0.10$, and a numeric cell endpoint required at least two of three replicate calls to agree. No qualifying endpoint and no consensus were retained as separate states. All activity fractions used the 27 experimentally tested peptides as the denominator. We examined strain-specific concentration distributions and threshold-dependent counts; the full threshold series and paired MIC values appear in Tables S5b–S5g. The algorithm was retrospective and did not replace the original experimental summary.

Results

Categorical reconstruction improves residue recovery

CE21 reconstructed 497 of 701 development residues (70.90%), compared with 88 (12.55%) for MSE21 and 111 (15.83%) for the scalar pipeline (**Table 1**). The CE21–MSE21 difference was 58.35 percentage points under matched architecture, initialization, schedule and decoder. CE21 recovered the correct length for all 41 sequences and reproduced three complete sequences. Its amino-acid macro recall was 42.76%; per-residue coverage is reported in Table S1.

Direct generation evaluation distinguishes the models

All eight method–terminal groups produced 16 unique canonical sequences of length 1–48, with no exact match to a training sequence. CE21 had lower values than both MSE21 and the scalar pipeline for all six distributional distances in both terminal conditions and against both reference sets. This gives 24 consistently improved distance comparisons for the categorical objective.

The LSTM had the lowest value for every specified distance in these comparisons (**Figure 1**). Development MMD^2 was 0.017328 for CE21 and 0.011567 for the LSTM in the free condition, and 0.007710 and 0.007362, respectively, in the amidated condition. The small amidated difference is reported descriptively. These results show that reconstruction accuracy and direct generation evaluation provide distinct information for model selection.

Complementary diagnostics refine this interpretation. CE21 produced 14 and 15 sequences of length 5–48 in the free and amidated groups, compared with 16 and 16 for the scalar pipeline and LSTM. Scalar outputs also had smaller absolute training-entropy discrepancies than CE21 in both terminal groups. The consistent advantage of CE21 therefore applies to the six specified distances; complete eligibility, entropy and residue-coverage results are given in Tables S1–S2.

Latent analyses separate encoded information from sampling performance

Latent-conditioned autoregression tested whether the encoded representation contained recoverable sequence information (**Figure 2**). Its eight-draw reconstruction score was 0.690802 with matched latents, compared with 0.097533 after latent mismatch and 0.091958 for the control trained with permuted latent–sequence pairs. An additional capacity-unmatched comparator scored 0.120436 and is reported descriptively. Nevertheless, that route did not satisfy its generation criterion. Encoded sequence information was therefore detectable without a corresponding improvement under the sampling criterion.

Training-posterior substitution did not improve development MMD, coordinate permutation worsened it, and the template-conditioned prior produced mixed results. KL warmup reduced the posterior-mean reconstruction measure from 0.722796 to 0.657852. Direct-latent diffusion and conditioning-context variants did not satisfy their respective generation-related criteria. Together, the seven analyses in Table S4 locate useful targets for further optimization; they do not identify a single causal explanation for the reconstruction–generation difference.

Source-group validation of activity regression

Activity regression examined whether sequence-based predictors improved quantitative MIC estimates over a simple reference. On the $\log_{10}$ MIC scale, primary MAE for the mean, ridge and CNN predictors was 0.811036, 0.775373 and 0.819758 for mass-unit records, and

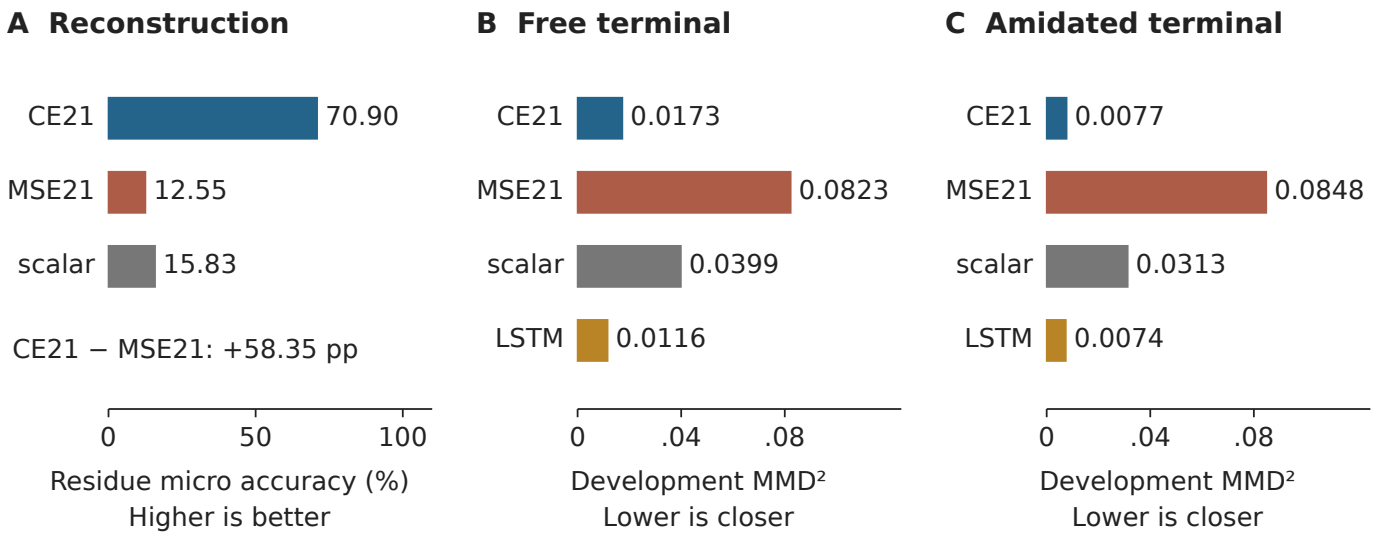


Figure 1 Reconstruction and generation under the fixed development protocol. **(A)** Development residue micro accuracy for the three reconstruction paths. **(B, C)** Development fixed-feature MMD^2 for the four generators in free and amidated terminal conditions; lower values indicate closer feature means. CE21 improves on MSE21, while the LSTM is closest on this generation metric. The scalar pipeline is not a matched single-component control. Results use one fitted seed, 701 reconstruction residues and 16 generation requests per method–terminal group; no uncertainty interval is inferred. Full numerical results are provided in Tables S1–S2.

Alt text: Three panels compare reconstruction accuracy and development MMD squared. CE21 reaches 70.90% reconstruction accuracy versus 12.55% for MSE21. LSTM has the lowest generation MMD squared in both terminal conditions.

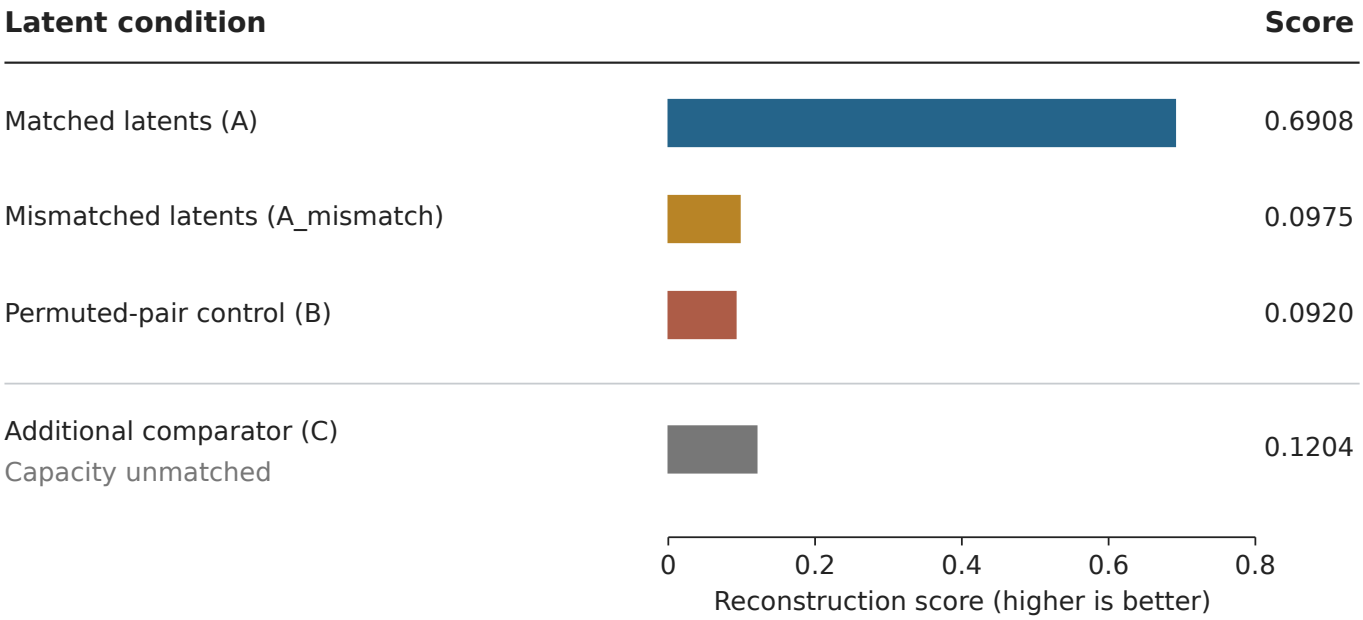


Figure 2 Latent-conditioned reconstruction under explicit controls. Matched latents, mismatched latents and a control trained with permuted latent–sequence pairing yield eight-draw reconstruction scores of 0.690802, 0.097533 and 0.091958, respectively. The additional comparator C scores 0.120436; its capacity is unmatched, so that contrast is descriptive. These measurements show recoverable encoded sequence information. The same route did not satisfy its six generation-distance comparisons; the figure does not depict a generation improvement. No seed-based uncertainty interval is inferred. Complete definitions and all seven exploratory analyses are provided in Table S4.

Alt text: Four horizontal bars show a reconstruction score of 0.690802 for matched latents and scores from 0.091958 to 0.120436 for the three controls; the fourth comparator has unmatched capacity.

0.516993, 0.582247 and 0.576262 for molar-unit records. The CNN did not improve on the mean under the specified source-group evaluation. Full weighting, censoring and variance analyses are presented in Table S3. These predictors were not used to select the experimental panel.

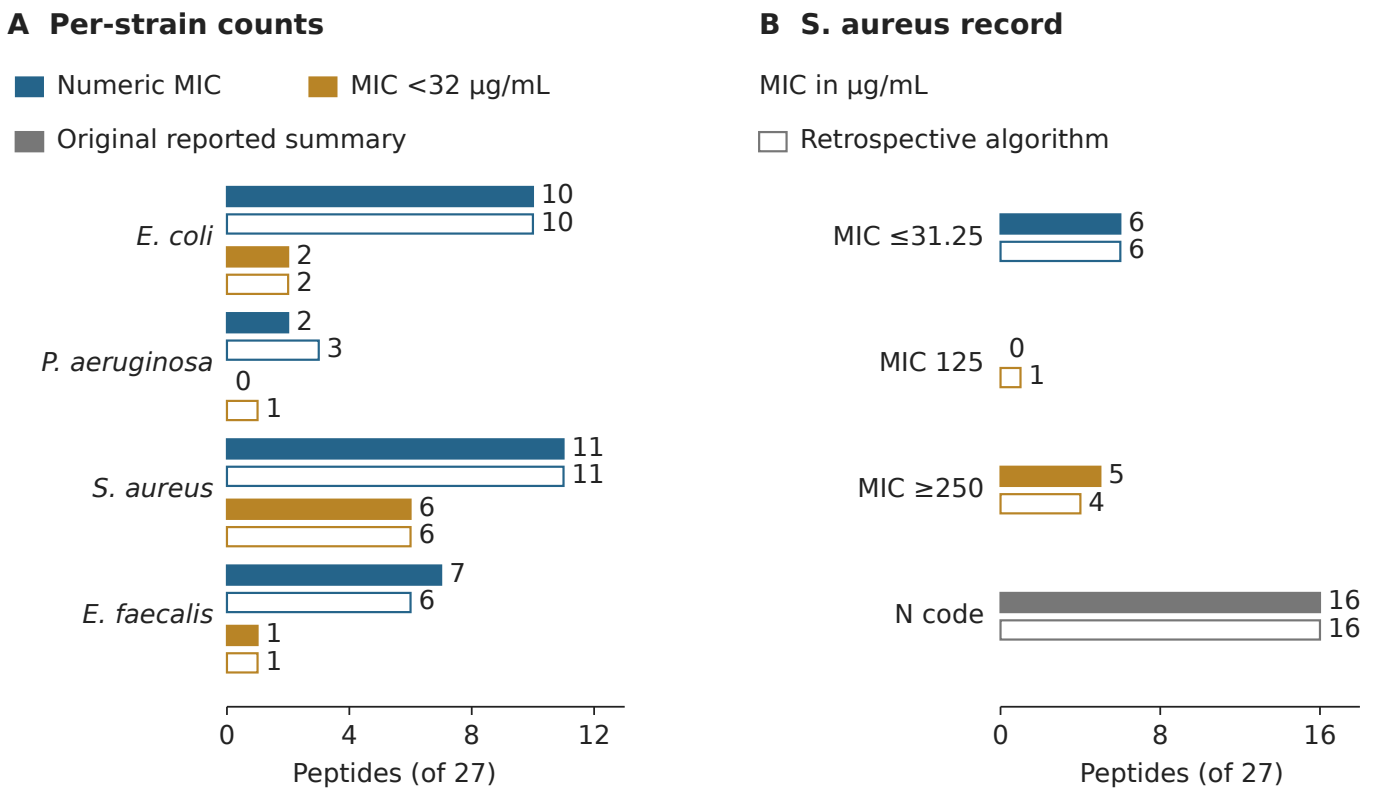


Figure 3 Antibacterial activity of 27 tested peptides, with original and replicate-based MIC estimates. **(A)** Numeric MIC counts and counts strictly below 32 µg/mL for each strain, with both representations shown. **(B)** *S. aureus* distributions: the reported summary contains six values at or below 31.25, five at or above 250 µg/mL and 16 N values; the algorithm contains six, four and 16, respectively, plus one value at 125 µg/mL. N combines majority-no-endpoint and no-consensus states only for historical comparison. Six peptides have *S. aureus* MICs ≤31.25 µg/mL under both representations; complete paired values are in Table S5.

Alt text: Filled and outlined bars compare reported and algorithm endpoints by strain. The *S. aureus* distribution changes from five values at or above 250 to four, with one value at 125; its count below 200 changes from six to seven.

Antibacterial activity in the 27-peptide panel

Six of the 27 tested peptides inhibited *S. aureus* ATCC 29213 at MICs ≤31.25 µg/mL (22.2%). These were pp1-3, pp1-5, pp2-8, pp3-11, pp3-13 and pp3-14. Two peptides, pp1-5 and pp3-13, had MICs of 7.8125 µg/mL against this strain. Peptide pp1-5 also showed MICs of 31.25, 125 and 125 µg/mL against *E. coli*, *P. aeruginosa* and *E. faecalis*, respectively, demonstrating activity across the four-strain panel. Across all 27 peptides, seven had MICs below 200 µg/mL against at least one strain (25.9%).

Both the six-peptide *S. aureus* count at ≤31.25 µg/mL and the seven-peptide any-strain count at <200 µg/mL were unchanged in the replicate-based sensitivity analysis (**Figure 3**). The original and algorithmic estimates differed in three of 108 cells (Table S5c); at the broader *S. aureus* threshold of <200 µg/mL, they yielded 6/27 and 7/27, respectively. Table S5 reports every peptide, including nonnumeric outcomes, and retains both MIC representations.

Discussion

Interpreting reconstruction and generation together

The controlled objective comparison identifies a practical improvement: categorical reconstruction increased residue recovery by 58.35 percentage points and improved all six specified generation distances relative to squared error. Direct comparison with the recurrent model then added a distinct conclusion: the LSTM had the lowest values on all 24 specified generation-distance comparisons. Both observations are needed to interpret the objective change. Reconstruction measures recovery with sequence-derived information, whereas generation measures sampling from a terminal condition alone.

The latent-conditioned experiment reinforces this distinction. Its matched-versus-mismatched contrast demonstrates recoverable sequence information, while the sampling result concerns how that representation is used during generation. Sequence VAEs in other domains also distinguish latent-variable use from successful generation (42). This is a methodological analogy: loss scaling, effective KL regularization, latent-distribution mismatch and decoder optimization remain plausible contributors here. The present experiments identify the endpoints requiring optimization rather than isolating one of these

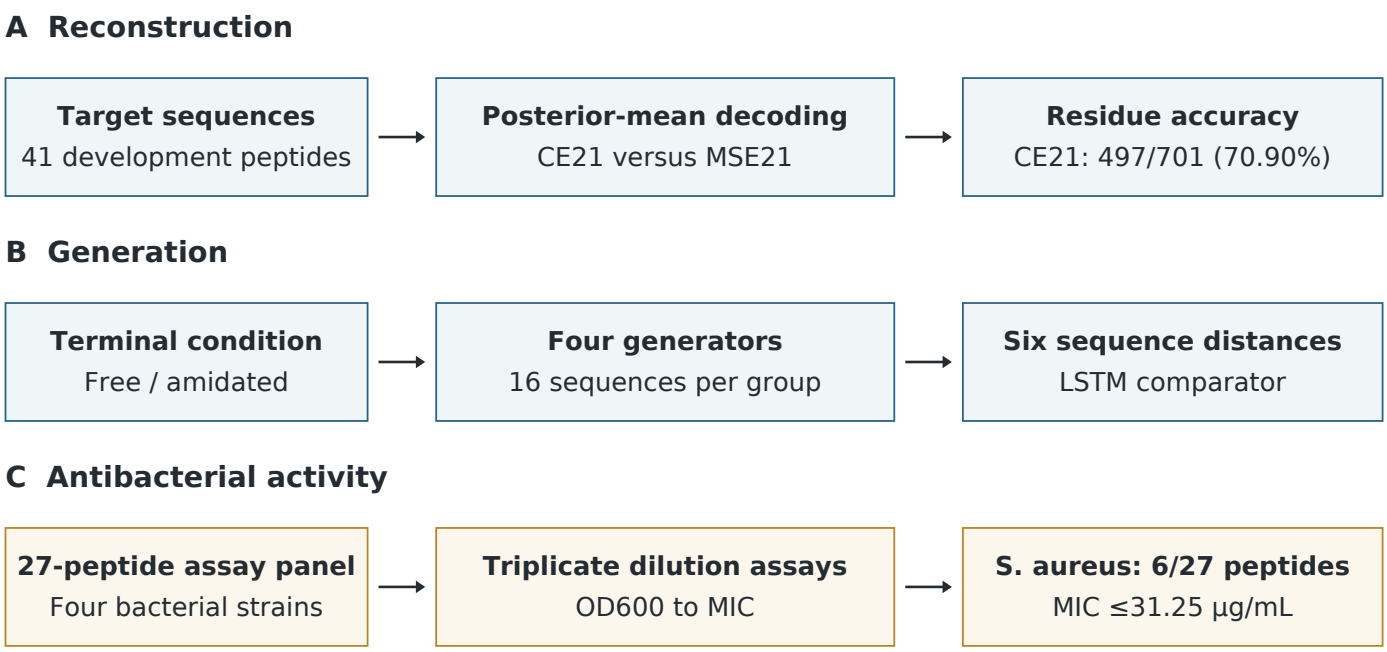


Figure 4 Endpoint-specific evaluation of antimicrobial peptide design. The three rows map the input and measurement to the corresponding result: reconstruction of known sequences, direct evaluation of sampled sequences and antibacterial assessment of a separate experimental panel. Within-row arrows indicate evaluation steps; the experimental panel is not a downstream output of the CE21–MSE21 comparison. Reconstruction uses 701 residues from 41 development sequences; generation uses 16 requests per method and terminal condition; activity uses 27 tested peptides.

Alt text: Three parallel rows connect reconstruction inputs to residue recovery, generated sequences to distributional distances, and a 27-peptide assay panel to strain-specific MIC results.

mechanisms. This evaluation complements AMP classification studies that use explicit controls to clarify model performance (22; 23) and general biological benchmarking standards (26; 28).

Connecting sequence design with antibacterial assessment

The 27-peptide panel identifies experimentally active candidates and resolves their concentration-dependent activity across four reference strains. The six *S. aureus*-active peptides at $\leq$31.25 µg/mL provide a concrete result that is stable across the two MIC analyses. Retaining sample-level measurements also permits sensitivity analysis without treating a nonnumeric entry as a measured zero.

AMPGen provides a published example of the complementary generation–selection–assay workflow (9). Its initial two-target activity assessment was followed by testing a selected subset on a broader strain panel. This staged design illustrates why a generation study should state the candidate-selection step and the biological endpoint it measures. The separate 27-peptide panel in this study likewise characterizes strain-specific activity; it is not a head-to-head assay of CE21 and MSE21 outputs. Work on non-haemolytic peptide design measures antibacterial and haemolytic activity separately (43), illustrating how an additional functional endpoint can extend a sequence-design study without being inferred from antibacterial activity alone.

An endpoint-specific evaluation protocol

The combined analyses motivate three linked evaluation decisions (**Figure 4**). First, an objective comparison should hold architecture, inputs and decoding fixed and quantify reconstruction at both residue and whole-sequence levels. Second, sampled sequences should be assessed directly against explicit reference groups and an executable comparator, with length, diversity and copy diagnostics accompanying distributional distances. Third, experimental activity should be reported per peptide, strain and concentration, with the tested panel as denominator and replicate analysis available alongside the primary MIC summary.

The framework assigns each observation to the scientific question it answers. Model-specific laboratory validation additionally pairs each tested peptide with its generating model and sampling configuration. This pairing supports future end-to-end evaluations while preserving the separate interpretation of the computational comparison and experimental panel presented here. Detailed implementation and endpoint definitions are available in the supplementary material.

The data representation should preserve the annotations needed to interpret these endpoints. APD3 organizes experimentally characterized antimicrobial peptides (44); CAMPR4 includes natural and synthetic sequences with modification information (45); DRAMP 4.0 expands annotations relevant to stability and safety (46). Their complementary scope illustrates why sequence identity, terminal chemistry and biological assay labels should remain explicit. The computational cohort analyzed here uses DBAASP, as specified in Methods.

Scope and future evaluation

The computational comparison uses one fitted seed, 16 samples per terminal group and a development partition used during model development. Its rankings therefore describe this operating point. Multiple seeds, an independent evaluation set and matched tuning budgets would test their stability. Holding the KL coefficient fixed does not equalize effective regularization across differently scaled reconstruction losses, and the compared model families differ in capacity and realized computation. The follow-up analyses help prioritize these next experiments.

Antibacterial MICs characterize inhibition under the tested panel; haemolysis, cytotoxicity and in vivo efficacy require separate evaluation (47; 48). A prospective study linking generated sequences to a fixed sampling protocol and predefined assays would extend the present endpoint-specific evaluation to model-level biological validation.

Conclusion

Categorical reconstruction improved residue recovery and six generation distances over a matched squared-error control, while direct generation evaluation identified the recurrent comparator as lowest on the six specified distances to the reference groups. In a separate 27-peptide panel, six peptides had *S. aureus* MICs $\leq$31.25 µg/mL and seven inhibited at least one tested strain below 200 µg/mL. Evaluating reconstruction, generation and antibacterial activity with their own explicit endpoints provides an interpretable basis for advancing antimicrobial peptide design.

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Author contributions

Hongsheng Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

Fei Wu: Conceptualization (academic supervision), Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

Feng Lin: Resources; Conceptualization (industrial scenario and requirements); Writing – Review & Editing; Supervision (industrial supervisor).

Fei Wu is the guarantor of the work.

Competing interests

Hongsheng Wang's affiliation with Loctek Ergonomic Technology Corp. and the institutional collaborations are disclosed in the author list. The authors report no competing interests beyond these disclosed collaborations.

Relationship to prior work

Hongsheng Wang and Feng Lin are also authors of AMPGen (9). The present work evaluates reconstruction and generation endpoints and reports a separate experimental peptide panel. AMPGen is cited as methodological context for staged candidate generation and biological assessment.

Ethics statement

No human participants, human tissue or animal subjects were involved in this work. All bacterial work used the standard reference strains described in Materials and methods.

Data and code availability

The versioned data and code are available at `https://github.com/wanghongsheng01/bib-amp-review-materials-20261023/tree/v5-2026-10-23`. The repository includes model implementations, configurations, checkpoints, saved computational outputs, peptide sequences, OD_{600} measurements, paired MIC estimates and supplementary tables. Offline analysis scripts reproduce the reported summaries. Supplementary Methods identify source versions and explain the computational and assay calculations.

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The authors wrote the manuscript. OpenAI Codex assisted with source and reference checking, code inspection, independent recalculation of saved outputs, deterministic plotting code, factual corrections and language editing of the author-written text. No new laboratory measurements or generative images were created. Human authors retain responsibility for the scientific content. Supplementary Methods record the assistance used in this revision.

Supplementary material

Supplementary Methods and Tables S1–S6 provide reconstruction and generation metrics, activity-regression analyses, latent analyses, complete peptide-level MIC data and context for staged biological evaluation. Source data and executable calculation scripts accompany the tables.

Key Points

- Categorical reconstruction increases development residue accuracy from 12.55% to 70.90% under a matched architecture and decoding protocol.
- CE21 improves six generation distances over MSE21; direct sampling evaluation identifies a terminal-conditioned LSTM as lowest on all 24 specified distance comparisons.
- Six of 27 tested peptides inhibit *S. aureus* at MICs $\leq$31.25 µg/mL; seven inhibit at least one of four strains below 200 µg/mL.
- An endpoint-specific protocol distinguishes reconstruction, sampled-sequence properties and measured antibacterial activity, with complete data supporting each result.

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